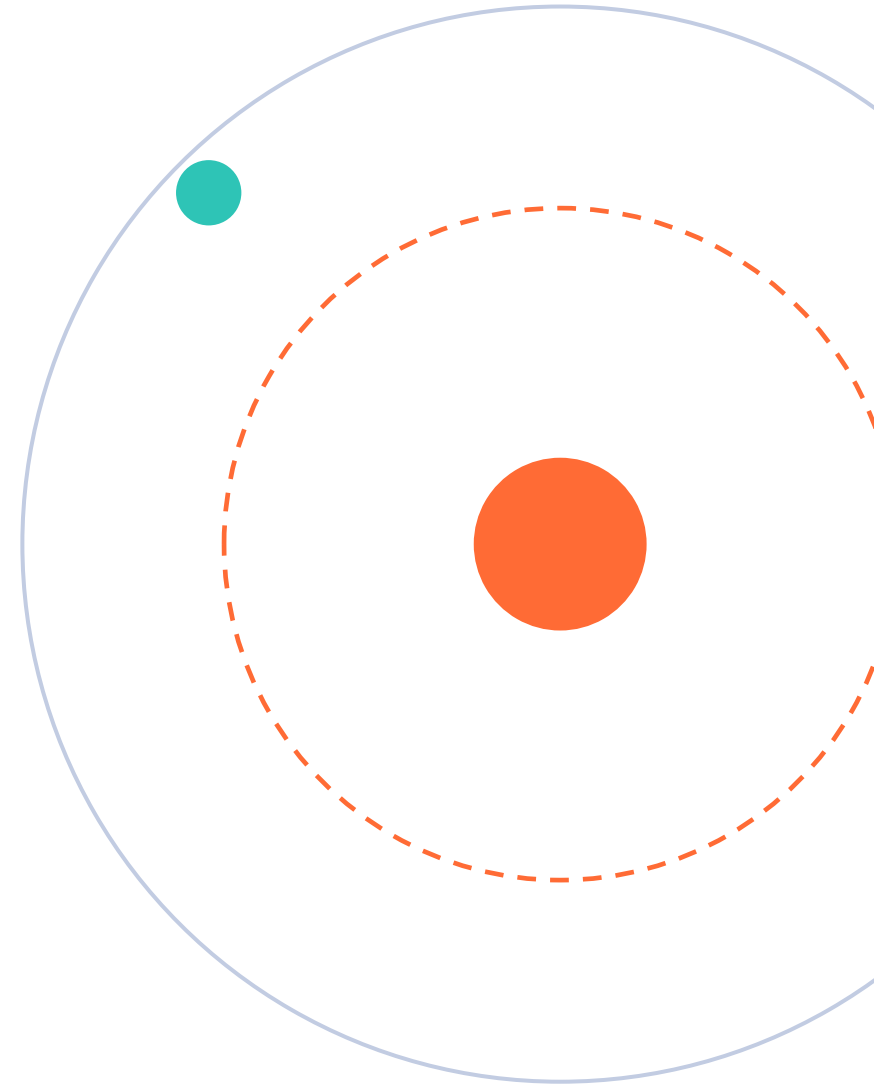


Doing Physics in the Age of AI

Wladimir E. Banda-Barragán

Universidad Yachay Tech



Where I come from

01

Computational astrophysicist

I simulate gas, shocks and turbulence.

02

First I learned to program badly

Then I found out how it should really be learned.

03

I teach physics WITH computing

From undergraduate to graduate level.

04

I have watched AI go from useless to sophisticated

From 2022 to 2026.

How NOT to learn to program

The language used in my first programming course

My first language: C

```
FILE *f = fopen("datos.dat", "r");  
double *x = malloc(N*sizeof(double));  
for (i = 0; i < N; i++)  
    fscanf(f, "%lf", &x[i]);  
/* mean? fit? plot? */  
free(x); fclose(f);
```

C is not suited to data analysis.

No dynamic arrays, no plots, no fits.



Too abstract

Pointers and memory before any physics.



No context

Nobody said what we were computing, or what for.



Purely mechanical

Edit the pattern, make it run, forget it.

How it is really learned

The order matters a great deal



My role as a lecturer: what I teach

Undergraduate



Quantum Mechanics

Formalism, operators



Computational Physics I

Python — numerics, ODEs, plotting



Computational Physics II

Python + HPC — PDEs, parallelism

Master's



Computational Fluid Dynamics

Euler and Navier–Stokes, solvers

NEW



Nonlinear Dynamics and Chaos

Maps, attractors, bifurcations



Quantum Mechanics for Nanoscience

Applied quantum mechanics, modelling

Teaching programming that sticks

Concepts first, then problems, projects and research goals

1

COMPETENCE

Can you solve a problem you have never seen?

Not “do you remember the command?”. Competence transfers; memory does not.

2

SYNTAX LAST

Syntax is vocabulary, not the language

Teach the idea and let the syntax attach itself to something meaningful.

3

PROBLEM SOLVING

Every programming exercise is a real physics problem

Decompose, estimate, code, check against the physics.
In that order.

4

PROJECTS AND RESEARCH

Real scientific goals

An open question worth asking. Ownership teaches by itself.

The test: take the tool away and see what is left.

Does an AI agent do physics yet?

ChatGPT • Claude • Gemini
Copilot • DeepSeek • Grok

Language models: they predict the next piece of text from enormous corpora of text and code.

Since 2022 I have fed them my own exam problems, plus textbook and homework problems.



My own investigation

My teaching experience, not a controlled benchmark.



Two things are tested

Physical reasoning and programming competence (Python).



New problems every semester

Designed by me.

AI solving my problems, 2022–2026

% solved correctly · graded by me

Simple prompt

The problem statement and nothing else

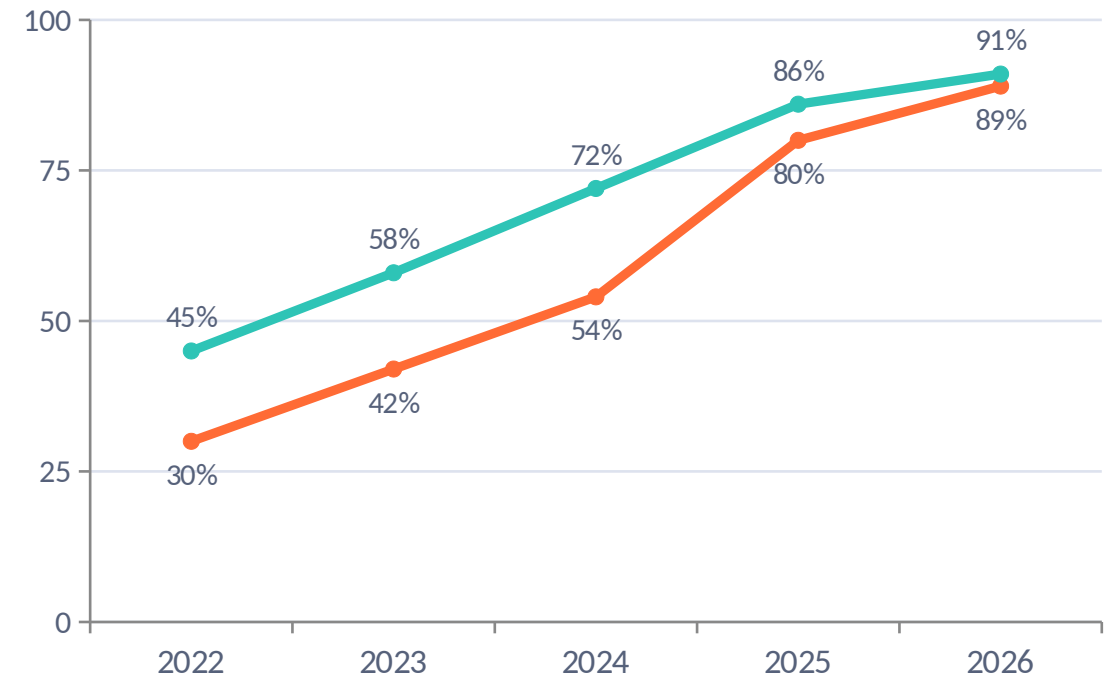


Advanced physics problems

Python / scientific code

Sophisticated prompt

Context, guidance and iteration



The difference between the two panels is the user, not the AI agent.

Orders of magnitude consistent with Kortemeyer, Phys. Rev. Phys. Educ. Res. 19, 010132 (2023) and with MATH, Hendrycks et al. (2021).

Year-by-year evolution

Years	Physics	Programming in Python
2022–2023	Basic arithmetic failures, wrong linear algebra, invented formulas. Misleading interpretation.	Toy scripts only. Hallucinated libraries. It fell over on numerics and on multi-file work.
2024	Real improvement from feedback training. The set-up is usually right.	Working code, but far too complex. Reasonable NumPy/SciPy. Debugging still weak.
2025	The jump: reliable mathematics, and it can solve advanced MIT-level problems for the first time.	It writes, runs and debugs complete code workflows. In paid versions it can automate outflows if the software architecture is supplied.
2026	Correct derivations, but the inference is still poor and the interpretation shallow.	Vectorization and HPC patterns work. Clean code, but not scalable (multi-prompts + a lot of time = good results).

- **Introductory physics:** ChatGPT would barely pass a calculus-based course. *Kortemeyer, Phys. Rev. Phys. Educ. Res. 19, 010132 (2023)*
- **Competition mathematics:** from $\approx 40\text{--}50\%$ (GPT-3) to more than 80% (GPT-4). *MATH, Hendrycks et al., arXiv:2103.03874 (2021)*
- **Research physics:** 5.7% for the best frontier model; 12.6% with tools. *CritPt, arXiv:2509.26574 (2025)*

2025: the great AI jump

Two advances, and one limit that did not move



The mathematics became reliable

Not “almost always right”. Reliably right.



The derivations improved

With a simple prompt they still carry errors and random assumptions; with a sophisticated prompt they improve considerably.



Interpretation did not advance

Standardised, shallow arguments; summaries vague in content.

THE UNCOMFORTABLE PART

The same AI agent can be a magnificent study assistant, and it can equally be the perfect way never to learn physics or acquire real competence in programming.

What it did to my classroom

Three eras

2022-23



A bad idea

They used it anyway. It was wrong so often that using it cost them marks.

2024-25



The sweet spot

Enough to help, not enough to replace the work by hand. My best teaching and learning outcomes.

2026



It got away from us

9 out of 10 assignments now read as AI-generated. Not assisted: generated.

The best year for learning was the year the tool was mediocre.

What I see in 2026

> 9 / 10

students already use AI to complete their
physics and physics-programming assignments.
The use is inevitable.

Getting stuck is part of learning

Learning happens in the twenty minutes of being stuck.

Debugging your own code is doing physics

You find your own mistakes because you know what the answer should look like.

Assessment stopped measuring competence

Written assessment does not match the assignment results. The same phrasing; tidy code, but with the same structure.

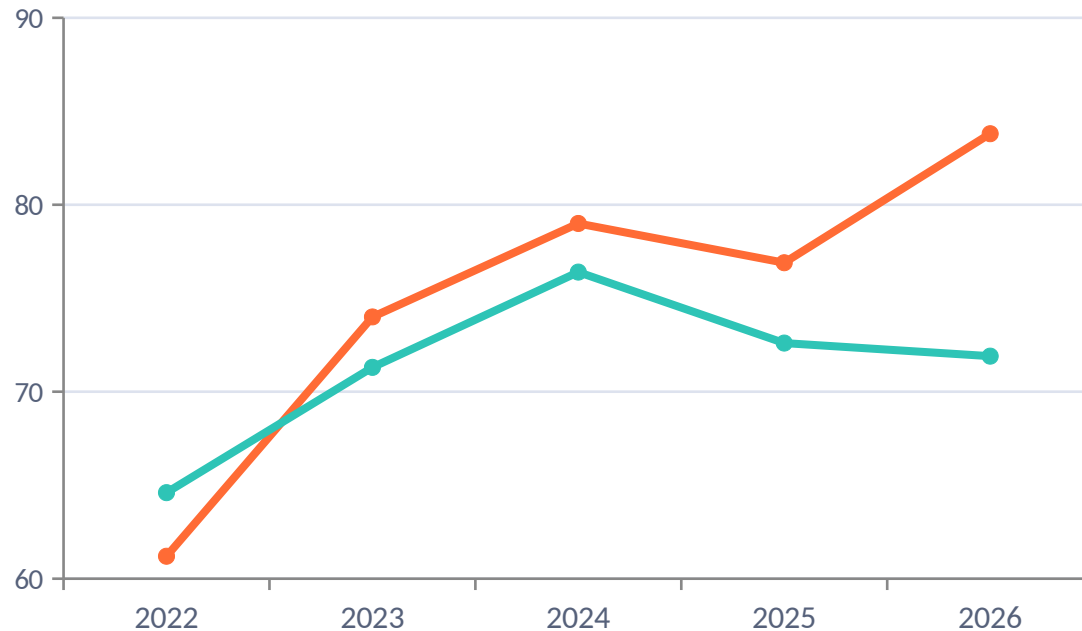
Confidence without competence

Correct work, no ability whatsoever to defend it. If several students ask the same AI agent, they hand in the same code.

Competence and proficiency metrics by course

% out of 100 • coding assignments, in-class work and exams

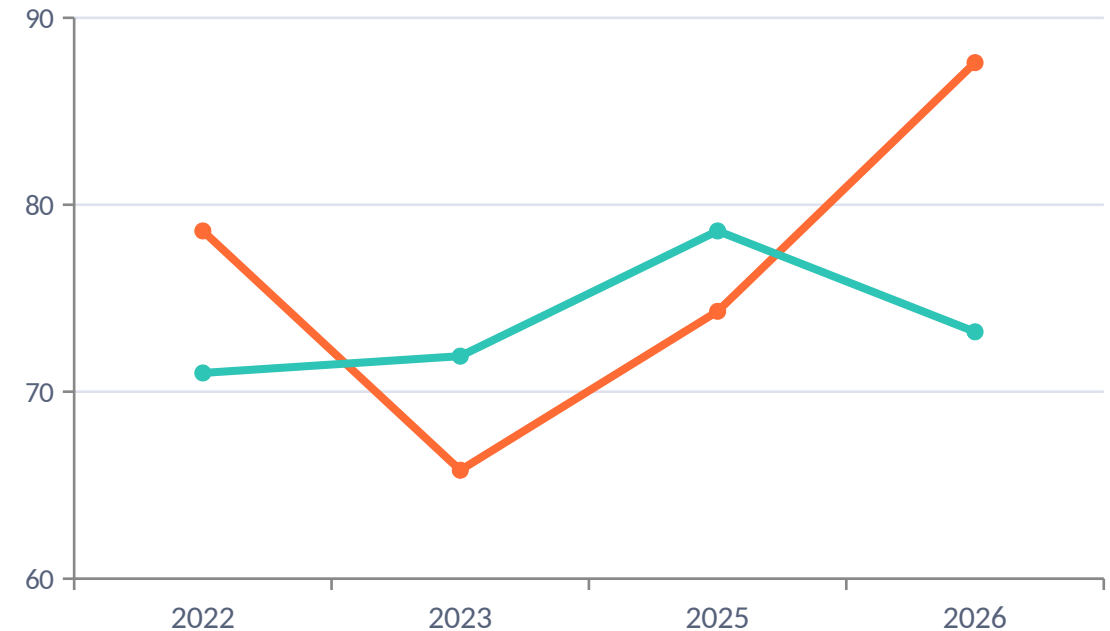
Computational Physics I



2026 gap: +12 pts

■ Coding assignments — at home
 ■ Closed-book, no-Internet work — in class

Computational Physics II



2026 gap: +14 pts

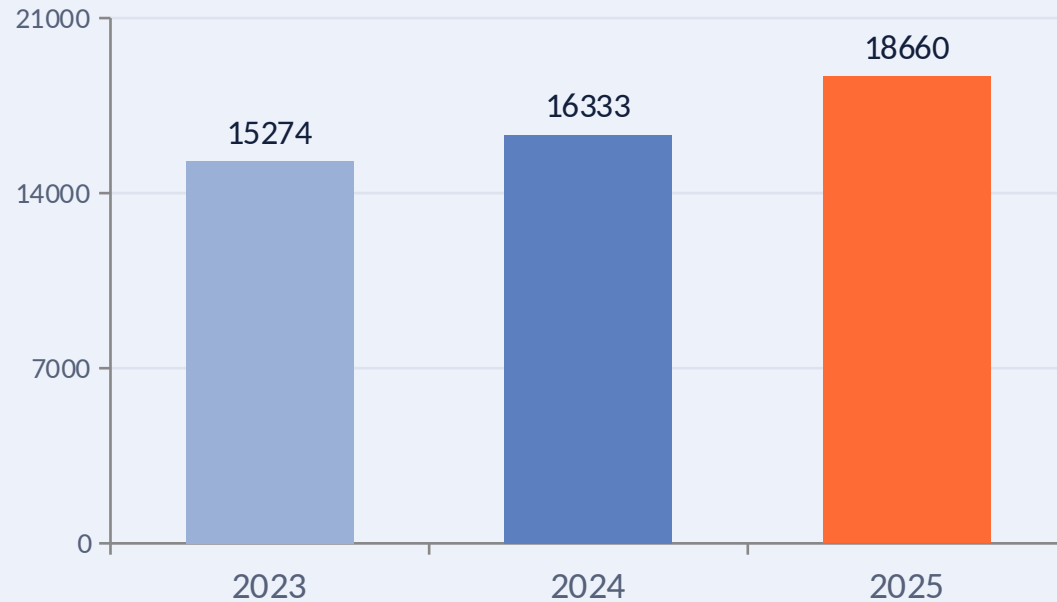
In both courses the lines separate in the last year.

My own records: Computational Physics I n=81 • Computational Physics II n=57.

And in research, what is happening?

More manuscripts than ever; models that still do not do frontier physics

New papers in astro-ph (arXiv)



+14 % in a single year: the largest jump of the decade.

CritPt benchmark (2025) · 71 unpublished research problems

5.7 %

the best frontier model, unaided

12.6 %

the same model with a calculator
and web search



It solves exercises, not questions

An exercise has a known answer. Research does not.



It fails on the complete problem

20–25 % on subtasks. On the complete problem, it falls apart.



It invents references

Plausible citations that do not exist. If you have not read it, do not cite it.

CritPt: a benchmark of 71 unpublished research problems written by more than 50 working physicists, arXiv:2509.26574 (2025). Chart: arXiv.

Why do we do physics?

Hogg (2026) puts the underlying question: if the AI agent can do it, what are we doing?



Practising and innovating is doing science

Every project that counts as physics involves novelty: measuring, interpreting or predicting something that had not been done. You do not learn it by reading or by taking classes: you have to do it.



Without trust there is no science

It is impossible to verify everything: what is not verified is trusted. With much of the literature in doubt, the work becomes unworkable.

“Let them cook”

Accept all of it: let AI conceive, execute, write and publish.

Hogg projects 10^8 – 10^9 times the human rate ($\times 10^5$ per project, $\times 10^4$ instances): coexistence is not possible. And if we delegate the work, we stop being physicists.

“Ban and punish”

Forbid AI and penalise anyone who uses it.

It forces a detection arms race that cannot be won, and it violates academic freedom, above all that of the youngest.

Hogg argues against both. There is a middle road, but finding it — and getting the community to agree on it — will not be easy.

What is fine, what is grey and what is not

AI use in physics research (the sciences): my view

✓ This is fine

- Searching the literature, verifying every citation.
- Having it explain a paper or a formalism you have ALREADY read in a paper.
- Help debugging YOUR own code, or speeding up a section of it.
- Polishing the wording of a text you have ALREADY written.

~ Grey zone — risky

- Drafting sections from results that really are yours.
- Derivations that you then check by hand, step by step. Risk of wrong mathematics.
- Summarising literature for an introduction. Risk of hallucinations and misreadings.
- Always declare it: using an AI model and not saying so is unacceptable.

✗ Not this

- Letting it write your code, your paper or your thesis.
- Putting a confidential manuscript under review — or your own — into a chatbot.
- Citing references you have neither read nor verified.
- Listing it as a co-author: it CANNOT take responsibility.

Springer Nature, Elsevier, AAAS, IEEE, PLOS and Taylor & Francis forbid AI in peer review; no journal accepts it as an author.

Problems ahead

What I think is coming if we do not act



The verification gap

They generate results but cannot tell a correct one from a plausible one. In research, that is fatal.



Low-level work dies first

Nobody learns C, Fortran or MPI if Python-plus-AI is enough. HPC astrophysics needs those people.



An assessment arms race

The detectors do not work. We cannot assess our way out of this.



The confidence trap

They believe they understand because the result looks like understanding.



Programming for theses

AI-generated codes with no user inputs, with certain bugs and wrong science.



...but also: a gift

Used well, it is the best tutor physics has ever had.

So what do you do?

Four rules for staying competent

1

Try first, ask afterwards

Twenty honest minutes on the problem before opening a model.

2

Use it as a tutor, not as a ghostwriter

“Explain why this is wrong” builds you up. “Do this for me” hollows you out.

3

Learn the mathematics behind it and do not use black boxes

If you cannot justify it and you do not understand it, do NOT use it — and it is NOT your work.

4

Learn one programming language properly

That is how you will see the real advantages of AI, and judge it better.

How I use AI myself

01 University bureaucracy

02 Web design in HTML

03 Speeding up and improving presentations (such as this one)

04 Code documentation and automation, on an architecture of my own

I do not use it for research or for scientific writing.

Conclusions

01

I learned to program the wrong way.

02

I taught myself, and found out how to program and how to teach programming well.

03

AI should help us with repetitive, tedious tasks.

04

In teaching, the future is to teach and to assess judgement in its use.

Questions?

Thank you, Claude!

